Supporting information

Draft of 2026-09-24

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Material that belongs to the record rather than to the argument: data provenance, protocol history, the predictions that failed, and the reproduction path. Nothing here is required to follow the article.

S1. Data sources

Daily water-temperature series were assembled from nine public providers. All are open archives; none required a data-use agreement.

provider	country	role in this study
ARSO	Slovenia	development
CHMI	Czechia	development
FOEN	Switzerland	development and reserved
GKD Bayern	Germany (Bavaria)	development and reserved
LUBW	Germany (Baden-Württemberg)	development and reserved
NVE	Norway	development and reserved
Rijkswaterstaat	Netherlands	development and reserved
SYKE	Finland	development and reserved
USGS	United States	development and reserved

Station coordinates were assembled for all 457 stations: from the providers' own catalogues already held in the repository for FOEN, CHMI, SYKE, LUBW, GKD Bayern, NVE and Rijkswaterstaat and for nine USGS gauges; from the USGS NWIS site service for the remaining 195 United States gauges; and from ARSO's public `hidro_podatki_zadnji.xml` feed for all 47 Slovenian stations. Every row in `data/interim/v18_station_coordinates.csv` records its source.

Daily 2 m air temperature is NASA POWER (MERRA-2), 1 January 1981 to 31 August 2026, one point per station at the product's native resolution: 16,679 days for each

of the 457 stations. The v17 programme used a much smaller forcing table covering 2.3% of development cases; both are retained and §5.7 compares them.

Basin characteristics for the United States subset are from GAGES-II (Falcone, 2011), joined on the eight-digit gauge identifier and matching 140 of 204 eligible stations.

S2. Quality control and the definition of a network

A *network* is a provider's group of stations on a common river system, which is the unit at which donor stations are available to one another. Networks are the clustering unit for every interval in the article, because station-years within a network share weather, regulation and instrumentation.

Waterbody class was assigned per network from the provider's own naming and an explicit map; seas, lakes, canals, estuaries and deltas are listed and excluded rather than averaged into a river result. Reference series are never imputed: a calendar year with any missing day contributes no truth. Stations with fewer than four fully observed calendar years are excluded, because the leave-one-year-out estimator protocol requires them.

The resulting denominator, with every exclusion and its count, is in `results/v17_fitness/step1_panel/`. The same stations contribute 2,422 station-years with at least 360 daily slots and at least one missing day; those form the deployment panel of §5.10 and appear in `results/v18_closure/e2_archive_deployment/real_station_`

S3. Protocol history

The study was pre-registered in `results/v17_fitness/step9_freeze/SUCCESS_CRITERION_v17.js` before the reserved networks were opened on 2026-09-23. Three aspects of that freeze were incomplete or wrong and are recorded here rather than silently repaired.

The incumbent's imputed limit was mis-stated. v17 placed the comparison at a thirteen-week (91-day) limit attributed to Johnson et al. (2021). The source says twenty weeks. `frontier.py` now uses 140 days and every artifact that used the old value was re-run; §5.6 reports the corrected operating point. The direction of the error flattered the incumbent at that point, so correcting it reduces the screen's advantage there from 6.2 to 4.4 percentage points.

The weighting between outage arms was not declared. The case set mixes an empirical arm, drawn from the outage geometry the same networks experience, with a stratified arm built to span the decision space. Their base rates differ by an order of magnitude. The pre-registered test was run on the pooled set, as declared; §5.9 reports the decomposition and states that the declared effect is not met on the empirical arm alone.

The natural scale of the peak has two values. Computed over the full eligible roster it is 1.06 °C; recomputed inside the development case table it is 1.08 °C. The primary tolerance is 0.50 °C throughout except in cross-metric comparisons, where each endpoint's scale is recomputed on the same footing. The tolerance sweep shows the axis ordering is unchanged between 0.25 and 1.0 standard deviations.

S4. Predictions that failed

Three predictions were written down before the analyses that tested them and were not met. All three are in the article; they are gathered here because a reader looking for them should not have to hunt.

1. **The support-restricted advantage will decay monotonically in the support width.** It does not: it is a hump peaking at a 60-day support. (§5.3)
2. **The estimand's own support is the best conditioning set at every support width.** It is not: below 30 days a fixed 30-day window beats it. (§5.3)
3. **For the sine-fit phase, two hand-chosen transition windows will beat both peak-window overlap and outage length.** They beat peak overlap decisively and tie outage length. (§5.3)

Failures 1 and 2 have one cause, which we then measured: the support's location is itself uncertain, so the conditioning set has to be the support dilated by that uncertainty. Failure 3 has a related cause: a hand-drawn window is a crude approximation to an influence function that is spread over half the year.

A fourth statement was **withdrawn** rather than falsified. v17 framed the standing guidance as getting an ordering backwards. On the guidance's own estimands it does not; §5.4 reports that a completeness threshold is the correct axis for the sine-fit mean and within 0.02 AUC of the best for the phase.

S5. Runs that were discarded

Two defects made specific v17 artifacts unusable and both were repaired before the reserved panel was opened: outage realizations were seeded from a value that Python randomises per process, and worker results were concatenated in completion order. A third affected the calibration rather than the data: an additive, per-estimator conformal scheme under-covered because selecting the smallest of several bounds is a winner's curse. All three are described in `paper/v17/SUPPLEMENT.md`; the affected numbers were discarded rather than reported alongside the repaired ones.

S6. Verification

- The case table is byte-reproducible including row order, pinned by a test.
- The v18 endpoint table regenerates the same 51,524 realizations from the same deterministic seeds, which is how its rows line up with the v17 case table.
- No feature may read a day the outage hides; enforced in `feature_frame` and pinned by a test.
- Every v18 table cell and audited inline number in the article is re-derived from the artifacts by `scripts/221_v18_audit_manuscript.py`, which the test suite runs; the article carries a machine-readable block of its headline numbers that is compared against those artifacts.
- Figure sources are `.drawio` XML and the PDFs are generated from them; a test checks that every schematic PDF is newer than its source.

S7. Reproduction

See `paper/v18/PROTOCOL.md`. Total wall time on 128 cores is about ninety minutes, of which the NASA POWER download is roughly twenty.